

Uniform Resurgent Crossover for Gaussian Binomial Coefficients

Exact remainders, a sharp modular-visibility threshold,
and certified inversion

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Abstract

The transseries companion in the ProveIt repository develops the singular $q \rightarrow 1$ crossover of central Gaussian coefficients, but explicitly leaves uniform matching at the classical endpoint and the finite-size tail's beyond-all-orders accuracy untreated. We address these questions for the specified analytic interpolation, rather than for arbitrary interpolations of the same sequence. A positive double-kernel representation supplies an exact signed remainder after every Bernoulli truncation. It yields an all-order expansion uniform for $x \rightarrow \infty$ and $0 \leq hx \leq T$, including the ordinary central-binomial endpoint. We compute the meromorphic Borel transform, its parity cancellations, and its imaginary-axis residue jump. The principal quantitative result is the uniform sharp estimate

$$\min_{K \geq 0} r_K(\tau/x, \tau) = \frac{e^{-2\pi x}}{\pi\sqrt{x}} (1 + O_T(x^{-1})), \quad 0 \leq \tau \leq T,$$

with every minimizing index $K = \pi x + O_T(1)$. Comparison with the exact modular correction $\mathcal{E}(h) \sim e^{-4\pi^2/h}$ gives a transition at $\tau = hx = 2\pi$, and a resolved boundary layer $\tau = 2\pi - (\log x)/(2x) + s/x$ in which the error-to-signal ratio tends to e^{-s}/π . This is a truncation-accuracy transition, not a Stokes transition. Finally, a uniform slope bound transfers residual certificates to inverse errors, and inversion about the exact Borel-summed carrier gives a convergent modular series with finite divisor-sum coefficients. Full proofs, reproducible numerical checks, a source audit, and further research questions are included. General q -Pochhammer resurgence is established prior work; the contribution claimed here is the explicit uniform, sharp-remainder, and inversion package, not priority for that general theory.

Status. Conventional mathematical proofs and exact symbolic checks; no Lean verification or peer review is claimed. Novelty relative to the inspected repository chapter is identified below. Literature-wide priority has not been established.

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1 Problem, source boundary, and principal contributions

1.1 The precise repository question

The repository snapshot used here is commit `04e06e032dff1966513bfba316966d52db3646b4` of ProveIt. The directly relevant source is the chapter *The singular transition $q \rightarrow 1$* in *Combinatorial Transseries and Their Inverses*; see [1, 2]. Its finite-order tail proposition `t3:prop:double-A` is uniform only for $\tau \geq \tau_0 > 0$. The discussion following `t3:eq:Ssmall` explicitly says that uniform matching at $\tau = 0$ requires another error analysis. The discussion of `t3:eq:modularsectors` and `t3:eq:central-double-asympt` separates an exact modular product from an algebraic remainder and does not claim to have resolved the finite-size tail’s entire beyond-all-orders contribution. Those are the two scoped questions developed in this paper.

We inspected the companion’s relevant source ranges, approximately lines 4090–4250, and the two group READMEs. The much larger canonical *Transseries and Inversion* source could not be retrieved through the available file endpoint. Its README and the group concordance were read, but no exhaustive nonduplication audit of that entire volume is claimed. The repository itself carefully distinguishes human mathematical statements from its narrower Lean crosswalk; this manuscript preserves that distinction.

The crossover already has the correct dilogarithmic leading phase, a positive phase derivative, formal higher inverse coefficients, and an exact eta-modular factor in the companion. These are starting points, not results claimed as discoveries here. The questions are quantitative: can one cross the singular endpoint uniformly, what is the *smallest actual* truncation error, and when can the much smaller modular contribution be resolved by that representation?

1.2 What is proved, and what is not

Our main results are as follows; all refer to the explicitly defined interpolation in section 2.

- (i) Theorems 3.1 and 4.1 give an exact positive remainder and an all-order expansion uniform on the closed crossover range $0 \leq hx \leq T$.
- (ii) Theorem 5.1 and proposition 5.2 give the explicit Borel pole divisor, its central parity cancellation, and its classical endpoint limit.
- (iii) Theorems 6.2 and 7.2 determine the optimal error including its leading constant and the logarithmically shifted modular-visibility boundary layer. The lower bound concerns all ordinary truncations, not only a suggested least-term index.
- (iv) Theorems 8.1 and 8.2 provide real residual certificates and a convergent modular inverse series about the exact, nonmodular carrier.

The proofs do not establish resurgence of every inverse formal series, a general Hahn-field construction, or a solution of the wider median-sum conjectures in the literature. Nor does a lower bound for ordinary truncations prohibit a more accurate integral, hyperasymptotic, or exact product algorithm. An exact positive remainder is itself a way around the truncation limitation.

1.3 Relation to established literature

Costin and Garoufalidis developed resurgence for Euler–Maclaurin summation [3]. Garoufalidis and Kashaev give an explicit meromorphic Borel transform, poles, residues, and Stokes behavior for Faddeev’s quantum dilogarithm [4]. Fantini and Rella study q -Pochhammer summability, quantum modularity, and character-weighted modular resurgent series; the version reviewed

here is dated 1 April 2026 [5]. Thus neither the use of Borel transforms for q -products nor the existence of their pole lattices is new in itself.

The elementary derivation below is included to fix all signs and normalizations and to make the central quotient's cancellations available for the uniform remainder theorem. The narrow contribution advanced for further expert checking is the assembled theorem with a uniform endpoint, the sharp $1/(\pi\sqrt{x})$ error constant, its precision boundary layer, and the explicitly certified inverse construction. The source comparison identifies a genuine extension of the inspected chapter; it is not a claim that no equivalent theorem exists elsewhere.

2 Normalization and the two small scales

For $h > 0$ put $Q = e^{-h}$ and define

$$(a; Q)_\infty = \prod_{j=0}^{\infty} (1 - aQ^j), \quad P(Q) = (Q; Q)_\infty. \quad (1)$$

Our real interpolation, for $x > 0$, is

$$H_h(x) = e^{hx^2} \frac{(Q^{x+1}; Q)_\infty^2}{P(Q)(Q^{2x+1}; Q)_\infty}, \quad F_h(x) = \log H_h(x). \quad (2)$$

For integer $n \geq 0$, it is exactly the central Gaussian coefficient $\left[\begin{smallmatrix} 2n \\ n \end{smallmatrix} \right]_q$ with $q = e^h$: the usual $q \leftrightarrow q^{-1}$ factor is q^{n^2} . All logarithms in real-variable formulas are real logarithms of positive quantities.

The standard q -gamma definition gives

$$H_h(x) = e^{hx^2} \frac{\Gamma_Q(2x+1)}{\Gamma_Q(x+1)^2}, \quad H_0(x) := \frac{\Gamma(2x+1)}{\Gamma(x+1)^2}. \quad (3)$$

The fixed- x limit as $h \downarrow 0$ is $H_0(x)$, by the q -gamma limit [7]. The parameter q here is continuous; the limit $q \rightarrow 1$ is not a limit through orders of finite fields.

Write $\tau = hx$. The tail and modular correction are

$$A(h, \tau) = \sum_{m \geq 1} \frac{e^{-m\tau}}{m(e^{hm} - 1)} = -\log(Qe^{-\tau}; Q)_\infty, \quad (4)$$

$$z(h) = e^{-4\pi^2/h}, \quad \mathcal{E}(h) = -\log P(z(h)) = \sum_{j \geq 1} \sigma_{-1}(j) z(h)^j, \quad (5)$$

$$\sigma_{-1}(j) = \sum_{d|j} \frac{1}{d}.$$

We set $\mathcal{E}(0) = 0$. Absolute convergence proves the divisor sum by expanding $-\log(1 - z^m)$ and grouping the product of the two indices. The eta transformation [8] yields

$$P(e^{-h}) = \sqrt{\frac{2\pi}{h}} \exp\left(-\frac{\pi^2}{6h} + \frac{h}{24}\right) P(z(h)). \quad (6)$$

Consequently, exactly,

$$F_h(x) = \frac{\tau^2 + \pi^2/6}{h} - \frac{1}{2} \log \frac{2\pi}{h} - \frac{h}{24} + A(h, 2\tau) - 2A(h, \tau) + \mathcal{E}(h). \quad (7)$$

The repository's phase and amplitude are

$$S(\tau) = \tau^2 + \frac{\pi^2}{6} + \text{Li}_2(e^{-2\tau}) - 2\text{Li}_2(e^{-\tau}), \quad (8)$$

$$J(\tau) = \frac{1}{2} \log(1 - e^{-2\tau}) - \log(1 - e^{-\tau}) = \frac{1}{2} \log \coth(\tau/2). \quad (9)$$

Differentiating the dilogarithms gives

$$S'(\tau) = 2\log(1 + e^\tau), \quad S''(\tau) = \frac{2e^\tau}{1 + e^\tau}, \quad S(0) = 0. \quad (10)$$

The phase S is strictly increasing. The two potentially competing exponentials in the crossover are $e^{-2\pi x}$ and $e^{-4\pi^2/h}$. The former will arise as a *minimal remainder size*; the latter is an actual, separately specified term in an exact identity. These interpretations must not be interchanged.

3 An exact positive remainder

3.1 From a partial fraction identity to a signed error

The standard partial fractions for $\coth$ [6] give, for $v > 0$,

$$\frac{1}{e^v - 1} = \frac{1}{v} - \frac{1}{2} + 2v \sum_{n \geq 1} \frac{1}{(2\pi n)^2 + v^2}. \quad (11)$$

The positive summands permit Tonelli's theorem in (4). Thus

$$A(h, \tau) = \frac{\text{Li}_2(e^{-\tau})}{h} + \frac{1}{2} \log(1 - e^{-\tau}) + 2h \sum_{m, n \geq 1} \frac{e^{-m\tau}}{(2\pi n)^2 + (mh)^2}. \quad (12)$$

Define

$$w_m(\tau) = 2e^{-m\tau} - e^{-2m\tau} > 0, \quad (13)$$

$$R(h, \tau) = 2h \sum_{m, n \geq 1} \frac{w_m(\tau)}{(2\pi n)^2 + (mh)^2}, \quad (14)$$

$$a_k(\tau) = \frac{|B_{2k}|}{(2k)!} \left(2\text{Li}_{2-2k}(e^{-\tau}) - \text{Li}_{2-2k}(e^{-2\tau}) \right) > 0. \quad (15)$$

Here $v/(e^v - 1) = \sum_{j \geq 0} B_j v^j / j!$ fixes the Bernoulli convention. The elementary equality $|B_{2k}|/(2k)! = 2\zeta(2k)/(2\pi)^{2k}$ can also be obtained by comparing Taylor coefficients in (11).

Theorem 3.1 (Exact signed remainder and enveloping bounds). *For every $h, \tau > 0$ and integer $K \geq 0$, put*

$$r_K(h, \tau) = 2h^{2K+1} \sum_{m, n \geq 1} \frac{w_m(\tau) m^{2K}}{(2\pi n)^{2K} ((2\pi n)^2 + (mh)^2)}. \quad (16)$$

Then

$$R(h, \tau) = \sum_{k=1}^K (-1)^{k-1} a_k(\tau) h^{2k-1} + (-1)^K r_K(h, \tau), \quad (17)$$

$$0 < r_K(h, \tau) < a_{K+1}(\tau) h^{2K+1}. \quad (18)$$

In particular, adjacent partial sums bracket R . The exact forward identity is

$$\begin{aligned} F_h(x) &= \frac{S(\tau)}{h} - \frac{1}{2} \log \frac{2\pi}{h} + J(\tau) - \frac{h}{24} \\ &\quad + \sum_{k=1}^K (-1)^k a_k(\tau) h^{2k-1} + \mathcal{E}(h) + (-1)^{K+1} r_K(h, \tau). \end{aligned} \quad (19)$$

Proof. For $a, b > 0$, finite geometric division gives

$$\frac{1}{a^2 + b^2} = \sum_{j=0}^{K-1} \frac{(-1)^j b^{2j}}{a^{2j+2}} + \frac{(-1)^K b^{2K}}{a^{2K}(a^2 + b^2)}.$$

Apply this with $a = 2\pi n$, $b = mh$, and multiply by $2hw_m$. Each coefficient sum is absolutely convergent: its m -part is an exponentially weighted polynomial, and its n -part is a zeta value of order at least two. Summing gives (17) and (16). Replacing the final denominator $a^2 + b^2$ by a^2 proves the strict upper bound (18). Positivity is immediate. Substituting (12) into (7) proves (19) with the displayed sign. $\square$

3.2 Why this is stronger than a formal Bernoulli expansion

A finite-order $O(h^{2K+1})$ estimate does not identify the sign, is not necessarily uniform as $\tau \downarrow 0$, and cannot justify choosing K of order $1/h$. In contrast, (16) is an exact identity for *every* K , with no restriction relating K, h, τ . We may therefore study the simultaneous limit $K \asymp x$ directly rather than substitute a moving K into a theorem proved only for fixed K .

There is an additional structural advantage. As a function of $z = h^2$, $R(h, \tau)/h$ is a positive Stieltjes-type sum of kernels $1/(a + bz)$. The resulting remainder sequence is a moment sequence.

Lemma 3.2 (Strict log-convexity in the truncation index). *For fixed $h, \tau > 0$,*

$$r_K^2 < r_{K-1} r_{K+1} \quad (K \geq 1). \quad (20)$$

The quotients r_{K+1}/r_K are strictly increasing, and $r_K \rightarrow \infty$ as $K \rightarrow \infty$. In particular, a minimum over $K \geq 0$ exists, with at most two adjacent minimizing indices.

Proof. Let

$$\nu_{m,n} = \frac{2hw_m}{(2\pi n)^2 + (mh)^2}, \quad t_{m,n} = \left(\frac{mh}{2\pi n} \right)^2.$$

Equation (16) says $r_K = \sum \nu_{m,n} t_{m,n}^K$. Cauchy–Schwarz applied to the two powers $K-1$ and $K+1$ gives (20); equality would force all $t_{m,n}$ with positive weight to coincide, which they do not. A single summand with $t_{m,n} > 1$ proves divergence. Increasing successive quotients prove the final claim. $\square$

4 Uniform closure of the classical endpoint

Define the normalized coefficient functions

$$\mathcal{A}(\tau) = \frac{S(\tau)}{\tau}, \quad \mathcal{A}(0) = 2 \log 2, \quad (21)$$

$$\mathcal{J}(\tau) = \frac{1}{2} \log \left(\frac{\tau}{2} \coth \frac{\tau}{2} \right), \quad \mathcal{J}(0) = 0, \quad (22)$$

$$D_k(\tau) = \frac{B_{2k} \tau^{2k-1}}{(2k)!} \left(\text{Li}_{2-2k}(e^{-2\tau}) - 2 \text{Li}_{2-2k}(e^{-\tau}) \right) - \mathbf{1}_{k=1} \frac{\tau}{24}. \quad (23)$$

The factor $-\tau/24$ must be kept in D_1 ; it comes from eta modularity, not from the finite-size kernel.

Theorem 4.1 (All orders uniformly through $\tau = 0$). *The functions in (21)–(23) extend real-analytically to $\tau = 0$. For $k \geq 1$,*

$$D_k(0) = \frac{B_{2k}}{2k(2k-1)} (2^{1-2k} - 2). \quad (24)$$

For every fixed $T > 0$ and integer $K \geq 0$, uniformly for $0 \leq \tau \leq T$ as $x \rightarrow \infty$,

$$F_{\tau/x}(x) = x\mathcal{A}(\tau) - \frac{1}{2} \log(\pi x) + \mathcal{J}(\tau) + \sum_{k=1}^K \frac{D_k(\tau)}{x^{2k-1}} + \mathcal{E}(\tau/x) + O_{T,K}(x^{-2K-1}). \quad (25)$$

For $K = 0$ the term $-\tau/(24x)$ may be absorbed in the stated remainder. For $K \geq 1$ the remainder is exactly $(-1)^{K+1}r_K(\tau/x, \tau)$. The endpoint interpretation is $F_0(x) = \log H_0(x)$ and $\mathcal{E}(0) = 0$.

Proof. The derivative (10) is analytic at zero, and $S(0) = 0$; therefore $S(\tau)/\tau$ has the asserted removable singularity. Also $(\tau/2) \coth(\tau/2)$ is a positive even analytic function with value one at zero, so (22) is analytic there.

For an integer $p \geq 0$,

$$\text{Li}_{-p}(e^{-\tau}) = \left(-\frac{d}{d\tau}\right)^p \frac{1}{e^\tau - 1}.$$

The Laurent expansion at zero has leading term $p!\tau^{-p-1}$; there are no other negative powers. Taking $p = 2k - 2$ in (23) proves removability and (24). The elementary simplification

$$D_1(\tau) = \frac{\tau}{24} \left(\coth \tau - 2 \coth \frac{\tau}{2} \right)$$

also displays the cancellation of its apparent linear term.

For $\tau > 0$, rearranging (19) with $h = \tau/x$ gives (25). Define

$$b_{K+1}(\tau) = \tau^{2K+1} a_{K+1}(\tau).$$

The same Laurent argument shows that b_{K+1} has a finite continuous limit at zero. Thus b_{K+1} is bounded on $[0, T]$, and (18) supplies the required uniform constant. For $K = 0$, include the bounded term $\tau/(24x)$ separately. For fixed x , pass to $\tau = 0$ using (3); alternatively, the Riemann-sum limit below passes directly in the exact kernel identity. $\square$

For example,

$$\begin{aligned} \mathcal{A}(\tau) &= 2 \log 2 + \frac{\tau}{2} + \frac{\tau^2}{12} - \frac{\tau^4}{480} + O(\tau^6), \\ D_1(0) &= -\frac{1}{8}, & D_2(0) &= \frac{1}{192}, & D_3(0) &= -\frac{1}{640}, \\ D_4(0) &= \frac{17}{14336}, & D_5(0) &= -\frac{31}{18432}, & D_6(0) &= \frac{691}{180224}. \end{aligned}$$

Hence the expansion reduces to the logarithmic central-binomial Stirling series at the endpoint, with the correct normalization $2x \log 2 - \frac{1}{2} \log(\pi x)$.

Proposition 4.2 (Continuous exact endpoint remainder). *For $x > 0$ and $p = 2K$, set $L = 2\pi x$. The continuous extension of $r_K(\tau/x, \tau)$ at $\tau = 0$ is*

$$r_K^{(0)}(x) = \frac{1}{2\pi^2 x L^p} \sum_{n \geq 1} \frac{1}{n^{p+2}} \int_0^\infty \frac{(2e^{-u} - e^{-2u})u^p}{1 + (u/(nL))^2} du. \quad (26)$$

It is strictly positive, satisfies the same first-omitted-term bound, and is strictly log-convex as a sequence in K .

Proof. After the change $u = m\tau$, (16) is the corresponding Riemann sum with mesh τ . The exponential decay gives an integrable dominating bound for each fixed K, x ; the n -sum is dominated by $\sum n^{-p-2}$. This proves the limit. The inequalities pass to the integral (strictness also follows inside it). Rewriting the integrand as a positive measure against $(u/(nL))^{2K}$ proves strict log-convexity exactly as in lemma 3.2. $\square$

Remark 4.3 (Scope of the endpoint assertion). The uniform estimate is a large- x statement for the whole closed range $0 \leq hx \leq T$. It does not require hx to remain bounded away from zero, and it is not merely a fixed- x continuity statement for q -gamma. Conversely, it does not assert uniformity as $T \rightarrow \infty$; the proof and optimal-truncation constants below allow dependence on T .

5 Borel geometry and the cancellation of even axial poles

5.1 Convention and an explicit meromorphic transform

For a formal series without constant term our convention is

$$\mathcal{B}\left[\sum_{j \geq 0} c_j h^{j+1}\right](\xi) = \sum_{j \geq 0} \frac{c_j \xi^j}{j!}, \quad \mathcal{L}_0 b(h) = \int_0^\infty e^{-\xi/h} b(\xi) d\xi. \quad (27)$$

Thus there is no extra $1/h$ in the Laplace integral. The perturbative correction in (19), apart from phase and amplitude, is

$$\tilde{P}(h, \tau) = -\frac{h}{24} + \sum_{k \geq 1} (-1)^k a_k(\tau) h^{2k-1}. \quad (28)$$

The word perturbative here includes its full divergent Bernoulli series; it excludes the separately prescribed modular term $\mathcal{E}(h)$.

Theorem 5.1 (Borel transform, pole divisor, and exact sum). *For fixed $\tau > 0$, define*

$$\beta(\xi; \tau) = \frac{1}{4\pi^2} \sum_{n \geq 1} \frac{1}{n^2} \left\{ \frac{1}{e^{\tau+i\xi/(2\pi n)} - 1} + \frac{1}{e^{\tau-i\xi/(2\pi n)} - 1} \right\}. \quad (29)$$

Then the Borel transform of (28) is the meromorphic function

$$\begin{aligned} B_\tau(\xi) &= -\frac{1}{24} + \beta(\xi; 2\tau) - 2\beta(\xi; \tau) \\ &= -\frac{1}{24} - \frac{1}{2\pi^2} \sum_{n \geq 1} \frac{1}{n^2} \left\{ \frac{1}{e^{\tau+i\xi/(2\pi n)} - \epsilon_n} + \frac{1}{e^{\tau-i\xi/(2\pi n)} - \epsilon_n} \right\}, \\ \epsilon_n &= \begin{cases} 1, & n \text{ odd}, \\ -1, & n \text{ even}. \end{cases} \end{aligned} \quad (30)$$

All poles are simple. With $\ell \in \mathbb{Z}$, they are

$$\begin{cases} 4\pi^2 n \ell \pm 2\pi i n \tau, & n \text{ odd}, \\ 2\pi^2 n(2\ell + 1) \pm 2\pi i n \tau, & n \text{ even}. \end{cases} \quad (31)$$

At an upper-half-plane pole the residue is $i/(\pi n)$ for odd n and $-i/(\pi n)$ for even n ; the lower residues have the opposite signs. In particular, the nearest poles are $\pm 2\pi i \tau$ and every even-indexed pure imaginary candidate has cancelled.

The series in (30) converges normally away from its poles. There are no poles on the positive real axis, and for all $h > 0$,

$$\mathcal{L}_0 B_\tau(h) = -\frac{h}{24} - R(h, \tau). \quad (32)$$

Consequently the phase, amplitude, and positive-direction Borel sum of the Bernoulli block reconstruct $F_h(x) - \mathcal{E}(h)$ exactly.

Proof. Expand the exponentials in (29) geometrically for real ξ . Their sum is

$$2 \sum_{m \geq 1} e^{-m\tau} \cos(m\xi/(2\pi n)).$$

The identity

$$\int_0^\infty e^{-\xi/h} \cos\left(\frac{m\xi}{2\pi n}\right) d\xi = \frac{h}{1 + (mh/(2\pi n))^2}$$

shows that its Laplace transform is the last double sum in (12). Interchanging sums and integral is legitimate because $\sum_n n^{-2} \sum_m e^{-m\tau} < \infty$ and the Laplace weight is integrable. This proves (32).

The Taylor expansion of the cosine gives the Borel transform of the Bernoulli coefficients. Equivalently, comparison with (11) identifies those coefficients directly. The nearest poles give a positive Taylor radius, so these coefficient identifications are analytic identities near zero, not merely manipulations of a divergent Taylor series in h .

To obtain the parity form, reindex the terms of $\beta(\xi; 2\tau)$ by $n = 2j$ and use

$$\frac{1}{e^{2u} - 1} = \frac{1}{2} \left(\frac{1}{e^u - 1} - \frac{1}{e^u + 1} \right).$$

Its even-index Bose denominator cancels against $-2\beta(\xi; \tau)$, leaving the stated alternating denominator. Solving for its zeros gives (31). At an upper odd pole the derivative of the denominator is $i/(2\pi n)$, and at an upper even pole it is $-i/(2\pi n)$; multiplication by $-1/(2\pi^2 n^2)$ gives the residues. The two signs of the imaginary part do not overlap, and different positive n have different imaginary parts. Thus no further pole collisions occur.

On a compact set in the ξ -plane, the large- n denominators tend uniformly to $e^\tau - \epsilon_n$, both bounded away from zero. The factor n^{-2} proves normal convergence. This also proves that the displayed pole list is complete. $\square$

5.2 A precisely oriented residue jump

Only odd n contribute on the positive imaginary axis. Let two admissible Laplace contours pass on opposite sides of this ray. Fix their oriented difference to enclose its poles by *positive* small circles, so that the residue theorem supplies $+2\pi i$ times each residue. In a common convergence sector put

$$\zeta = \exp(-2\pi i \tau/h), \quad |\zeta| < 1.$$

Then the perturbative contour difference is

$$\Delta P = 2\pi i \sum_{\substack{n \geq 1 \\ n \text{ odd}}} \frac{i}{\pi n} \zeta^n = -2 \sum_{n \text{ odd}} \frac{\zeta^n}{n} = \log \frac{1 - \zeta}{1 + \zeta}. \quad (33)$$

The logarithm is the branch analytic at $\zeta = 0$ with value zero there. One can first deform contours past finitely many poles and then pass to the absolutely convergent residue sum; reversing the orientation reverses the sign. The ray has a gap from the other rays in (31), so no undisclosed poles enter this deformation.

Equation (33) concerns the Borel sum of (28). It is not a claim that the original product and its eta factor possess an unrestricted common analytic continuation across $\Re h = 0$. Keeping this boundary avoids identifying a valid local residue calculation with a substantially stronger global continuation theorem.

5.3 The classical Borel transform emerges uniformly

Use the actual endpoint expansion variable $\varepsilon = 1/x$, so $h = \tau\varepsilon$. Its Borel transform is

$$\mathcal{B}_\tau(s) = \tau B_\tau(\tau s). \quad (34)$$

Proposition 5.2 (Scaled Borel endpoint). *As $\tau \downarrow 0$, locally uniformly away from the indicated poles,*

$$\mathcal{B}_\tau(s) \longrightarrow \mathcal{B}_0(s) = -4 \sum_{\substack{n \geq 1 \\ n \text{ odd}}} \frac{1}{s^2 + (2\pi n)^2}. \quad (35)$$

For every $T > 0$ and $s \geq 0$,

$$|\mathcal{B}_\tau(s)| \leq \frac{T}{24} + \frac{1}{6}, \quad 0 < \tau \leq T. \quad (36)$$

In particular, the positive-direction Laplace sums pass to the classical endpoint by dominated convergence. The nearest scaled poles are always $\pm 2\pi i$, explaining the scale that appears in optimal truncation.

Proof. For odd n , the two fractions multiplied by τ tend to $(1 + is/(2\pi n))^{-1} + (1 - is/(2\pi n))^{-1}$. For even n , both tend to zero. The odd expression, multiplied by $-1/(2\pi^2 n^2)$, is exactly the summand in (35). Uniform domination by Cn^{-2} on compact sets away from the limit poles proves local normal convergence. Poles with nonzero real part in (31), after division by τ , escape to infinity.

For real s , $|e^{\tau \pm i\tau s/(2\pi n)} - \epsilon_n| \geq e^\tau - 1$. Since $\tau/(e^\tau - 1) \leq 1$, summation yields $\tau/24 + \zeta(2)/\pi^2 \leq T/24 + 1/6$. The Laplace weight e^{-xs} is integrable, which proves the final convergence statement. $\square$

Remark 5.3 (The missing flat term is not a summation ambiguity). The positive real Borel ray has no singularity and its sum is unambiguous. Nevertheless the exact normalized answer is that sum plus $\mathcal{E}(h)$. The formal power series alone cannot determine an independently added flat function. Here the product normalization and eta identity determine it. This example is therefore a useful concrete distinction between uniqueness of a chosen Borel sum and uniqueness among all functions sharing the same Poincaré expansion.

6 Sharp optimal truncation, including the leading constant

The nearest Borel singularity suggests the exponential size, but does not by itself prove the actual minimum or its prefactor. We prove both from the positive remainder.

6.1 A uniform discrete saddle estimate

We use the notation

$$p = 2K, \quad L = 2\pi x, \quad g_{n,L}(u) = \frac{1}{1 + (u/(nL))^2}. \quad (37)$$

A direct rescaling of (16) gives

$$r_K(\tau/x, \tau) = \frac{1}{2\pi^2 x L^p} \sum_{n \geq 1} n^{-p-2} \tau \sum_{m \geq 1} (2e^{-m\tau} - e^{-2m\tau})(m\tau)^p g_{n,L}(m\tau). \quad (38)$$

At $\tau = 0$ the inner Riemann sum means its integral as in (26).

Lemma 6.1 (Uniform gamma-saddle approximation). *Fix $T > 0$ and $0 < c < C < \infty$. As $p \rightarrow \infty$, uniformly for $0 \leq \tau \leq T$ and $c \leq p/L \leq C$,*

$$r_K(\tau/x, \tau) = \frac{p!}{\pi^2 x L^p (1 + (p/L)^2)} \left(1 + O_{T,c,C}(p^{-1})\right). \quad (39)$$

Proof. We first justify a relative, not merely absolute, Riemann-sum estimate. For a twice differentiable, decaying function whose endpoint derivatives and endpoint values vanish, the first periodic-Bernoulli remainder in the trapezoidal formula gives

$$\left| \tau \sum_{m \geq 1} f(m\tau) - \int_0^\infty f(u) du \right| \leq \frac{\tau^2}{12} \int_0^\infty |f''(u)| du. \quad (40)$$

For completeness this follows by applying the finite-interval Euler–Maclaurin identity through the B_2 term, letting its upper endpoint go to infinity, and using $\sup_{0 \leq t \leq 1} |B_2(t)| = 1/6$. Here $f(u) = f_0(u)g_{1,L}(u)$ and $f_0(u) = e^{-u}u^p$, with $p > 2$, so all the required endpoint terms vanish.

Normalize f_0 by $p!$ and let U have this gamma density. Since $f'_0/f_0 = p/u - 1$, the exact inverse moments of U give

$$\int_0^\infty |f'_0(u)| du \leq \frac{p!}{\sqrt{p-1}}, \quad (41)$$

$$\int_0^\infty |f''_0(u)| du \leq \frac{2p!}{p-1}. \quad (42)$$

Indeed, $\mathbb{E}(p/U - 1)^2 = 1/(p-1)$ and $\mathbb{E}(p/U^2) = 1/(p-1)$; Cauchy–Schwarz and the formula for f''_0 prove the bounds. The derivatives of $g_{1,L}$ satisfy $\|g'_{1,L}\|_\infty = O(L^{-1})$ and $\|g''_{1,L}\|_\infty = O(L^{-2})$. The product rule, with $L \asymp p$, therefore gives

$$\int |f''| \leq C_1 \frac{p!}{p}. \quad (43)$$

The integral $\int f = p! \mathbb{E}g_{1,L}(U)$ is bounded below by a positive constant times $p!$ when $p/L \in [c, C]$: for example, the event $U \leq 2(p+1)$ has probability at least $1/2$ by Markov’s inequality, and $g_{1,L}$ has a uniform positive lower bound on that event. Equations (40) and (43) prove a relative $O_T(p^{-1})$ error, also uniformly as $\tau \downarrow 0$.

The $n \geq 2$ contribution in (38) is $O(2^{-p})$ relative to the $n = 1$ contribution: discard its kernel by $g_{n,L} \leq 1$, use the just-established upper moment bound, and sum $\sum_{n \geq 2} n^{-p-2} = O(2^{-p})$. The negative exponential e^{-2u} contributes only $O_T(2^{-p})$ relatively. To see this without assuming a fixed mesh, put $v = 2u$ and apply the same Riemann estimate at mesh $2\tau \leq 2T$; its unweighted integral is $p!/2^{p+1}$. Hence (38) reduces to

$$\frac{1}{\pi^2 x L^p} \int_0^\infty \frac{e^{-u}u^p}{1 + (u/L)^2} du \left(1 + O_T(p^{-1})\right). \quad (44)$$

Finally, $\mathbb{E}U = p+1$ and $\text{Var } U = p+1$. Taylor’s theorem applied to $g_{1,L}$ about p gives

$$\mathbb{E}g_{1,L}(U) = g_{1,L}(p) + O(L^{-1}) + O((p+2)L^{-2}) = \frac{1}{1 + (p/L)^2} + O(p^{-1}).$$

The leading value is bounded away from zero on $[c, C]$, so this is a relative estimate. Substitution into (44) proves (39). The integral case $\tau = 0$ is included throughout. $\square$

6.2 Locating the true minimum

Theorem 6.2 (Uniform sharp minimum). *For every $T > 0$, uniformly for $0 \leq \tau \leq T$ as $x \rightarrow \infty$,*

$$\min_{K \geq 0} r_K(\tau/x, \tau) = \frac{e^{-2\pi x}}{\pi\sqrt{x}} \left(1 + O_T(x^{-1})\right). \quad (45)$$

Every minimizing index satisfies

$$K = \pi x + O_T(1). \quad (46)$$

More generally, the asymptotic expression in (45) holds for any integer $K = \pi x + O(1)$, uniformly when the implied displacement bound is fixed.

Proof. Take the quotient of (39) at $p+2$ and p . Uniformly when p/L stays in a fixed compact subinterval of $(0, \infty)$,

$$\begin{aligned} \frac{r_{K+1}}{r_K} &= \frac{(p+1)(p+2)}{L^2} \frac{1 + (p/L)^2}{1 + ((p+2)/L)^2} \left(1 + O_T(p^{-1})\right) \\ &= \left(\frac{p}{L}\right)^2 \left(1 + O_T(L^{-1})\right) \quad \text{when } p \asymp L. \end{aligned} \quad (47)$$

Choose a sufficiently large fixed C_T . At an even integer p within two of $L - C_T$, the quotient is less than one; at an even integer within two of $L + C_T$, it is greater than one. This follows by expanding $(1 \pm C_T/L)^2$, with C_T larger than the uniform constant in the error term. Strict log-convexity from lemma 3.2 and proposition 4.2 then puts every global minimum between these two indices. This is the step that upgrades a local least-term heuristic to a statement about all $K \geq 0$.

For $p = L + O_T(1)$, Stirling's formula gives

$$\frac{p!}{L^p} = \sqrt{2\pi L} e^{-L} \left(1 + O_T(L^{-1})\right). \quad (48)$$

Also $1 + (p/L)^2 = 2 + O_T(L^{-1})$. Since $\sqrt{2\pi L}/(2\pi^2 x) = 1/(\pi\sqrt{x})$, substitution into (39) proves (45). The same computation proves the assertion for any bounded displacement from πx . $\square$

Remark 6.3 (What fixes the universal constant). Three effects contribute. The $n = 1$ partial-fraction kernel dominates; $2e^{-u}$ dominates $2e^{-u} - e^{-2u}$ at the large-order saddle; and the kernel denominator is asymptotically $1 + (p/L)^2 = 2$. Its factor $1/2$ is essential. Replacing the exact remainder by the first omitted term would give a different prefactor and would not prove the sharp minimum. The finite mesh becomes invisible to relative order $O_T(1/x)$ because the gamma saddle has width of order $\sqrt{x}$ while $\tau \leq T$.

7 A sharp modular-visibility boundary

Lemma 7.1 (Elementary bounds for the modular series). *For $0 < z < 1$, writing $\mathcal{E}(z) = \sum_{j \geq 1} \sigma_{-1}(j)z^j$,*

$$z \leq \mathcal{E}(z) \leq \frac{z}{(1-z)^2}, \quad (49)$$

$$0 \leq \mathcal{E}(z) - \sum_{j=1}^M \sigma_{-1}(j)z^j \leq \frac{z^{M+1}((M+1) - Mz)}{(1-z)^2}. \quad (50)$$

In particular $\mathcal{E}(z) = z + O(z^2)$.

Proof. The coefficient is positive, equals one at $j = 1$, and satisfies $\sigma_{-1}(j) \leq j$. Sum $\sum jz^j$ and its tail. $\square$

The notation $\mathcal{E}(z)$ in this lemma is its generating function; elsewhere $\mathcal{E}(h)$ means its value at $z = e^{-4\pi^2/h}$.

Theorem 7.2 (Visibility threshold and resolved boundary layer). *Let $r_*(x, \tau) = \min_K r_K(\tau/x, \tau)$. For fixed $\tau > 0$,*

$$\frac{r_*(x, \tau)}{\mathcal{E}(\tau/x)} = \frac{1}{\pi\sqrt{x}} \exp\left[x\left(\frac{4\pi^2}{\tau} - 2\pi\right)\right] \left(1 + O_\tau(x^{-1})\right). \quad (51)$$

Thus the ratio tends to infinity for $0 < \tau < 2\pi$, to zero for $\tau > 2\pi$, and is asymptotic to $1/(\pi\sqrt{x})$ at $\tau = 2\pi$. For fixed real s , define

$$\tau_x(s) = 2\pi - \frac{\log x}{2x} + \frac{s}{x}. \quad (52)$$

Then, uniformly for s in compact subsets of $\mathbb{R}$,

$$\boxed{\frac{r_*(x, \tau_x(s))}{\mathcal{E}(\tau_x(s)/x)} \rightarrow \frac{e^{-s}}{\pi}}. \quad (53)$$

The same limit holds with any $K = \pi x + O(1)$ in place of the minimizer.

Proof. Combine (45) with (49). In the boundary layer put $\delta = \tau_x - 2\pi$. The exact expansion of the reciprocal gives

$$x\left(\frac{4\pi^2}{2\pi + \delta} - 2\pi\right) = -x\delta + O(x\delta^2) = \frac{1}{2}\log x - s + O\left(\frac{(\log x)^2}{x}\right).$$

The error is uniform for bounded s . Substitution proves (53); theorem 6.2 supplies the last assertion. $\square$

The equal-size boundary occurs, to this accuracy, at

$$\tau = 2\pi - \frac{\log x}{2x} - \frac{\log \pi}{x} + o(x^{-1}). \quad (54)$$

More generally, to make the best truncation residual a fraction $\eta > 0$ of the leading modular signal in the boundary scaling, the limiting condition is $s > -\log(\pi\eta)$. This is a quantitative precision budget, not merely a comparison of two exponentials with their polynomial prefactors discarded.

7.1 Interpretation and limitations

Suppose one approximates the nonmodular carrier by truncating its Bernoulli series, retaining phase and amplitude exactly. The smallest possible absolute forward error among such approximants is r_* . For $\tau < 2\pi$ fixed, every such truncation has error much larger than the modular correction. Simply appending a modular term to an algebraically truncated expression therefore does not resolve that term in absolute accuracy. At and above the threshold, near-optimal truncation does resolve it asymptotically.

This is not a positive-axis Stokes phenomenon: theorem 5.1 shows that there are no Borel poles on that axis. Nor is it a claim that a real exponential with action 2π must be inserted as an independent sector in the exact answer. It is the scale of the error left by optimal truncation of a Borel-summable block whose nearest poles are imaginary. Exact integration of that block, or direct evaluation of the positive remainder, can achieve accuracy below r_* . The theorem is sharp for its stated family of approximants, not for every conceivable algorithm.

8 Certified inversion and a convergent modular inverse series

All derivatives in this section are taken with h fixed. Although we use $\tau = hx$ to state uniform bounds, differentiating along the curve $h = \tau/x$ with τ fixed would be a different operation and would give the wrong inverse sensitivity.

8.1 A uniform lower bound for the exact slope

Theorem 8.1 (Monotonicity and quantitative inverse sensitivity). *For $h > 0$ and $x > 0$, the derivative is*

$$F'_h(x) = 2hx + 2h \sum_{m \geq 1} \frac{e^{-mhx} - e^{-2mhx}}{e^{hm} - 1} > 0. \quad (55)$$

Writing $\tau = hx$,

$$S'(\tau) - \frac{h}{2 \sinh \tau} \leq F'_h(x) \leq S'(\tau), \quad F'_h(x) \geq S'(\tau) - \frac{1}{2x}. \quad (56)$$

For $x \geq 1$ this gives the uniform positive constant

$$F'_h(x) \geq d_0 := 2 \log 2 - \frac{1}{2} > 0. \quad (57)$$

The bounds extend to $h = 0$ by continuity. If $F_h(x) = L$ and $x, \hat{x} \geq 1$, then

$$|x - \hat{x}| \leq \frac{|F_h(\hat{x}) - L|}{d_0}. \quad (58)$$

Proof. Differentiate the absolutely locally convergent series (7). Each summand in (55) is positive. For $v > 0$, (11) and $e^v \geq 1 + v$ imply

$$\frac{1}{v} - \frac{1}{2} \leq \frac{1}{e^v - 1} \leq \frac{1}{v}.$$

The upper comparison in (55) sums to $2\tau + 2 \log(1 + e^{-\tau}) = S'(\tau)$. The loss in the lower comparison is

$$h \sum_{m \geq 1} (e^{-m\tau} - e^{-2m\tau}) = \frac{h}{2 \sinh \tau} \leq \frac{1}{2x}.$$

This proves (56) and (57). For fixed x , the sum in (55) is a Riemann sum with mesh h for the continuous, integrable function $(e^{-xt} - e^{-2xt})/(e^t - 1)$. Its limit is $2 \int_0^\infty (e^{-xt} - e^{-2xt})/(e^t - 1) dt = 2\psi(2x+1) - 2\psi(x+1) = F'_0(x)$, which proves the endpoint derivative bound without differentiating a merely pointwise limit. Finally the mean value theorem on the interval between x and $\hat{x}$ proves (58). $\square$

A certificate should also certify existence, not merely an error conditional on a root. Suppose a computable approximation $T_h(t)$ has known bounds $|F_h(t) - T_h(t)| \leq \epsilon(t)$ at two points $1 \leq a < b$. If

$$T_h(a) + \epsilon(a) \leq L \leq T_h(b) - \epsilon(b), \quad (59)$$

continuity and strict monotonicity give one and only one root in $[a, b]$. At an interior trial point, (58) then gives

$$|x - \hat{x}| \leq \frac{|T_h(\hat{x}) - L| + \epsilon(\hat{x})}{d_0}.$$

The positive kernel and (50) supply such ϵ without making an asymptotic remainder into an unsupported numerical assertion.

8.2 Inverting the exact nonmodular carrier

Define

$$F_h^0(x) = F_h(x) - \mathcal{E}(h), \quad G_h^0 = (F_h^0)^{-1}. \quad (60)$$

By theorem 5.1, F_h^0 consists of the exact phase and amplitude and the Borel-summed Bernoulli block. It is not a finite truncation of that block. Since $\mathcal{E}(h)$ is independent of x , both carriers have the same derivative. If $x_0 = G_h^0(L)$, then the exact inverse relation is

$$F_h^{-1}(L) = G_h^0(L - \mathcal{E}(h)). \quad (61)$$

A shift in target therefore encodes every modular inverse sector.

Theorem 8.2 (Uniformly convergent modular inverse expansion). *Fix $T > 0$. Let $x_0 \geq 1$, $h > 0$, $hx_0 \leq T$, and $L = F_h^0(x_0)$. Put*

$$M_T = 2T + 8, \quad r = \min \left\{ \frac{1}{4}, \frac{d_0}{4M_T} \right\}, \quad \rho = \frac{d_0 r}{2}, \quad \mathcal{D}_h = \frac{1}{F_h'(x)} \frac{d}{dx}. \quad (62)$$

For $\mathcal{E}(h) < \rho$,

$$F_h^{-1}(L) = x_0 + \sum_{n \geq 1} \frac{(-1)^n \mathcal{E}(h)^n}{n!} \left[\mathcal{D}_h^{n-1} \frac{1}{F_h'(x)} \right]_{x=x_0}. \quad (63)$$

The series converges absolutely, and its tail after $n = N$ is at most

$$r \frac{(\mathcal{E}(h)/\rho)^{N+1}}{1 - \mathcal{E}(h)/\rho}. \quad (64)$$

It may also be collected into a convergent series in $z = e^{-4\pi^2/h}$:

$$F_h^{-1}(L) = x_0 + \sum_{m \geq 1} C_m(h, x_0) z^m, \quad (65)$$

where every coefficient is the finite expression

$$\begin{aligned} C_m(h, x_0) &= \sum_{n=1}^m \frac{(-1)^n}{n!} \left[\mathcal{D}_h^{n-1} \frac{1}{F_h'(x)} \right]_{x=x_0} \\ &\quad \times \sum_{\substack{j_1 + \dots + j_n = m \\ j_i \geq 1}} \prod_{i=1}^n \sigma_{-1}(j_i). \end{aligned} \quad (66)$$

If $0 < r_z < 1$ satisfies $r_z/(1 - r_z)^2 < \rho$, then for $z < r_z$ the tail of (65) after degree M is at most

$$r \frac{(z/r_z)^{M+1}}{1 - z/r_z}. \quad (67)$$

Proof. The product logarithms, hence F_h^0 , are holomorphic when $\Re x > 0$; the series provides a single-valued branch there. For $|w| \leq 1/4$, $\Re(x_0 + w) \geq 3/4$. Differentiating (55) once and estimating the result gives

$$\begin{aligned} |F_h''(x_0 + w)| &\leq 2h + 2h^2 \sum_{m \geq 1} \frac{m(e^{-mha} + 2e^{-2mha})}{e^{hm} - 1} \\ &\leq 2h + 2h \left(\frac{1}{e^{ha} - 1} + \frac{2}{e^{2ha} - 1} \right) \leq 2h + \frac{4}{a} \leq M_T, \end{aligned}$$

where $a = \Re(x_0 + w)$ and $h \leq T$ follows from $x_0 \geq 1$. Consequently, on $|w| = r$,

$$|F_h^0(x_0 + w) - L - F_h'(x_0)w| \leq \frac{1}{2}M_T r^2.$$

For $|s| < \rho$, the right side plus $|s|$ is strictly less than $d_0 r \leq |F_h'(x_0)w|$. Rouché's theorem gives a unique root of $F_h^0(x_0 + w) = L + s$ in $|w| < r$. Moreover $|F_h'(x_0 + w) - F_h'(x_0)| \leq M_T r \leq d_0/4$, so the derivative never vanishes there. The roots form one holomorphic inverse $G_h^0(L + s)$ on $|s| < \rho$, with $|G_h^0(L + s) - x_0| < r$.

Cauchy's coefficient bound is therefore $|[s^n](G_h^0(L + s) - x_0)| \leq r\rho^{-n}$. Repeated differentiation of an inverse gives

$$(G_h^0)^{(n)}(L) = \left[\mathcal{D}_h^{n-1}(1/F_h') \right]_{x_0} \quad (n \geq 1).$$

Taylor expansion at $s = -\mathcal{E}(h)$ proves (63), and the geometric sum gives (64).

For the collection into z , temporarily regard z as an independent complex variable, holding h, x_0 fixed in the carrier. By lemma 7.1, $|\sum \sigma_{-1}(j)z^j| < \rho$ for $|z| \leq r_z$. Absolute convergence of (63) and positivity of the coefficients of $\mathcal{E}(z)$ justify collecting powers at every positive z with $\mathcal{E}(z) < \rho$. On the indicated disk, the analytic composition is bounded by r relative to x_0 , and its coefficients are exactly (66) by the Cauchy product. A second Cauchy estimate gives (67). Evaluating the auxiliary variable at its physical value $z(h)$ completes the proof. $\square$

The first two coefficients illustrate the separate sources of nonlinear mixing:

$$C_1 = -\frac{1}{F_h'(x_0)}, \tag{68}$$

$$C_2 = -\frac{3}{2F_h'(x_0)} - \frac{F_h''(x_0)}{2(F_h'(x_0))^3}. \tag{69}$$

The $3/2$ is the divisor sum $\sigma_{-1}(2)$; the second term is inverse curvature. Suppressing either changes the second modular sector.

Uniformly for $hx_0 \leq T$ as $x_0 \rightarrow \infty$ and $h \downarrow 0$,

$$F_h^{-1}(L) - x_0 = -\frac{e^{-4\pi^2/h}}{S'(hx_0)} \left(1 + O_T(x_0^{-1}) + O_T(e^{-4\pi^2/h}) \right). \tag{70}$$

The shift is negative because adding a positive x -independent term to an increasing function lowers its inverse at a fixed target.

8.3 What the visibility theorem says for inverse accuracy

For a small forward residual δ , the exact local inverse displacement obeys

$$G_h^0(L + \delta) - x_0 = \frac{\delta}{F_h'(x_0)} + O_T(\delta^2). \tag{71}$$

In particular, compare the transported near-optimal residual of magnitude r_K with the modular target shift $\mathcal{E}(h)$, at the same center. Their inverse displacement ratio is

$$\frac{|G_h^0(L \pm r_K) - x_0|}{|G_h^0(L - \mathcal{E}(h)) - x_0|} = \frac{r_K}{\mathcal{E}(h)} (1 + O_T(r_K + \mathcal{E}(h))). \tag{72}$$

Thus theorem 7.2 persists under this exact local transport. This does not assert an optimal-truncation theorem for a separately constructed formal inverse series. Such a theorem would need its own large-order analysis. The convergent series in theorem 8.2 instead uses the *exact* carrier and so has a different, explicitly controlled source of error.

9 Reproducible computations and an executable error budget

9.1 Cancellation-free evaluation of a tiny remainder

Subtracting a truncated series from a function can destroy all significant digits at optimal truncation. The positive representation avoids that problem. With $p = 2K$, $s = p + 2$, and

$$c = \frac{2h^{p+1}}{(2\pi)^{p+2}}, \quad W_m = (2e^{-m\tau} - e^{-2m\tau})m^p,$$

it reads

$$r_K = c \sum_{m \geq 1} W_m \sum_{n \geq 1} \frac{n^{-s}}{1 + (mh/(2\pi n))^2}. \quad (73)$$

Let $R_{M,N}$ be the positive partial sum through $m = M$, $n = N$. If

$$\theta_M = e^{-\tau} \left(\frac{M+2}{M+1} \right)^p < 1,$$

then the exact omitted-tail estimate is

$$0 \leq r_K - R_{M,N} \leq c \left(\zeta(s, N+1) \sum_{m=1}^M W_m + \zeta(s) \frac{2e^{-(M+1)\tau} (M+1)^p}{1 - \theta_M} \right). \quad (74)$$

Here $\zeta(s, N+1) = \sum_{n > N} n^{-s}$ is the Hurwitz-zeta tail. To prove this, discard each kernel by one. The missing n -tail on retained m gives the first term. For omitted m , use $W_m \leq 2e^{-m\tau}m^p$; its successive ratios decrease once the first omitted index is reached, and are at most θ_M . Summing the resulting geometric majorant gives the second term. No subtraction of nearly equal large quantities occurs.

The supplied `verify.py` implements (73) and (74). It uses 110 decimal digits. Tail inequalities are proved in exact real arithmetic in this article; the program evaluates them with ordinary high-precision rounding, *not* outward-rounded interval arithmetic. Accordingly the reported runs are reproducible checks, not independently certified interval computations.

9.2 Actual results of the supplied run

The full script was run in preparing this manuscript. Exact symbolic arithmetic checks the six endpoint coefficients displayed after theorem 4.1. Eight exact identities also compare the endpoint Borel Taylor coefficients with the odd-pole lattice. Ten rational residue-cancellation checks are included. Independent finite products at integer indices check (7); the four absolute discrepancies were all below 3×10^{-110} . This is an independent normalization test because it does not evaluate both sides from the same infinite-product tail.

Table 1: Positive remainders checked against direct high-precision subtraction at nonoptimal orders.

x	τ	K	r_K	$r_K/(a_{K+1}h^{2K+1})$
5	1	4	$7.91517695222 \cdot 10^{-10}$	0.9191817471
7	2	6	$1.21204088308 \cdot 10^{-13}$	0.9160370429
8	0.5	6	$2.17819186412 \cdot 10^{-14}$	0.9340864521

Table 2: Near-optimal truncation at $\tau = 1$. The script uses $K = \text{round}(\pi x)$, not an unverified claim to have found the exact minimizing integer.

x	K	r_K	$r_K \pi \sqrt{x} e^{2\pi x}$
5	16	$3.17973793023 \cdot 10^{-15}$	0.9835348221
10	31	$5.21956022965 \cdot 10^{-29}$	1.0053351399
20	63	$1.88567218808 \cdot 10^{-56}$	0.9958300329
40	126	$3.55472089880 \cdot 10^{-111}$	0.9979153044

Each remainder in table 1 was positive, had the proved sign in the forward identity, and lay below the first omitted term. The positive partial-sum bounds enclosed the directly computed value at the stated working precision.

The mathematical relative upper bounds on omitted positive summands in these four evaluations, as evaluated by the program, are respectively below $4.2 \cdot 10^{-31}$, $6.1 \cdot 10^{-39}$, $1.3 \cdot 10^{-39}$, and $1.7 \cdot 10^{-38}$. This illustrates why a value below 10^{-110} can still be computed *relatively* at 110 digits: the computation uses positive small terms, not subtraction from a number of order one. At $x = 20$, tests with $\tau = 0.1$, 2π , and 10 give normalized ratios 0.9958300329, 0.9958300329, and 0.9958300327, respectively.

Table 3: Resolved boundary layer. The same near-optimal rule for K is used. Finite- x values need not approach their limits monotonically.

x	ratio at $s = 0$	ratio at $s = 1$
10	0.3269592979	0.1177674042
20	0.3227622917	0.1168425200
40	0.3220074088	0.1171886859
80	0.3214572071	0.1174603693
Limit	$1/\pi = 0.3183098862$	$e^{-1}/\pi = 0.1170996630$

The error in the exponent of (53) is of order $(\log x)^2/x$, so the visible approach in table 3 is slower than the $O(1/x)$ accuracy of the pure optimal-remainder formula.

For an inverse check, take $h = 1.2$, $x_0 = 5$, and the exact target $L = F_h(5) - \mathcal{E}(h)$. The independently computed inverse shift and its first-order prediction are

$$\begin{aligned} F_h^{-1}(L) - 5 &= -4.29546689467745006556 \cdot 10^{-16}, \\ -\mathcal{E}(h)/F'_h(5) &= -4.29546689467744988132 \cdot 10^{-16}. \end{aligned}$$

Their ratio is $1.00000000000000004289\dots$, with the correct negative sign. Separate slope checks at $(h, x) = (0.01, 1), (0.1, 20), (0.25, 4), (1, 3)$ satisfy both sides of (56).

9.3 Using the results rather than only inspecting coefficients

For a prescribed target L , a usable workflow is to choose a center from the repository's monotone dilogarithmic phase, bracket the exact root by (59), and then refine using the exact slope. A finite Bernoulli expansion is adequate when its first-omitted-term bound fits the error budget. Near exponential precision, evaluate the positive remainder or the exact Borel kernel and retain as many modular sectors as (50) requires. Equation (58) converts that forward budget into an inverse budget. The modular inverse series is useful when an exact carrier center is already available; it is not a replacement for obtaining that center accurately.

To reproduce the supplied computations, run

`python verify.py`

from the package directory. This writes `verification_results.json`. The optional `-quick` switch omits the larger cases. Python 3.10 or later, `mpmath`, and `SymPy` are required. The source was run with `mpmath` 1.3.0 and `SymPy` 1.14.0. The script raises an exception if an asserted test fails and uses no network service or proprietary computer algebra system.

10 Further research questions

The following questions separate extensions plausibly accessible from this kernel method from broader problems requiring additional ideas. They are proposed research targets, not claims that the corresponding statements are already proved.

10.1 Higher optimal-remainder coefficients and the minimizing integer

Question. Can one compute a complete expansion for $r_K \pi \sqrt{x} e^{2\pi x}$ in inverse powers of x , uniformly for bounded $K - \pi x$ and $0 \leq \tau \leq T$, and determine the exact minimizing integer outside an explicitly described tie set? The proof of lemma 6.1 keeps only a first relative error. Higher gamma moments and a sharper discrete-sum estimate should expose the integer-rounding dependence. It is important to distinguish a smooth large-order expansion from exponentially small mesh effects, particularly when two adjacent truncation orders nearly tie.

10.2 The range of a growing crossover parameter

Question. What is the largest growth range $T = T(x)$ for which the universal constant in (45) remains valid? The proof gives an initial diagnostic: the mesh error from (40) is $O(\tau^2/x)$ near the saddle. Thus $\tau = o(\sqrt{x})$ is a natural first range to investigate, but a sharp statement needs uniform constants and may be improved by exploiting analyticity rather than a twice-differentiable estimate. When the mesh is comparable to the saddle width, a genuinely discrete correction may remain. This is not covered by the fixed- T theorem.

10.3 General Gaussian quotients and positivity criteria

Question. For products of shifted q -gamma functions with balanced slopes, which combinations give a positive kernel analogous to $2e^{-u} - e^{-2u}$? What replaces the optimal error constant when the kernel changes sign? For positive kernels, moment log-convexity may survive and make a global minimum accessible. With sign-changing kernels, cancellation can defeat the present lower-bound argument even if the nearest Borel singularities are known. A useful classification should state both the slope data and the kernel-positivity condition, rather than infer one from the other.

10.4 Arithmetic descriptions of pole cancellations

Question. Can pole cancellation in finite products of q -Pochhammer factors be expressed as an explicit finite arithmetic transform of the slope and shift multiplicities? The central case reduces Bose denominators to alternating Bose/Fermi ones and removes even axial poles. Rational slope ratios suggest a finite residue-class calculation, whereas irrational slopes may create nonperiodic pole configurations. Establishing a criterion would turn a case-by-case cancellation into a reusable transseries operation.

10.5 Resurgence of the independently reverted crossover series

Question. Determine the Borel singularities and Stokes data of the formal inverse obtained by reverting the full polynomial–logarithmic crossover series, including its $\log h$ coefficients. The exact-carrier construction in theorem 8.2 does not answer this. Composition, moving singularities, and nonlinear sector mixing must be controlled. A satisfactory result should also identify which singularities are inherited from the forward kernel and which can arise from zeros of the analytically continued derivative.

10.6 A hyperasymptotic algorithm below the visibility threshold

Question. Can the positive remainder be expanded again, using the nearest pole pair, to produce a rigorously certified algorithm resolving $\mathcal{E}(h)$ for fixed $\tau < 2\pi$ with fewer operations than direct tail summation? There is no impossibility obstruction: exact kernel integration already contains the missing information. The question is computational complexity and stable error accounting. A useful result would specify the number of retained pole pairs and arithmetic precision as functions of h, τ and the requested relative accuracy of the modular term.

10.7 Root-of-unity and complex-parameter crossover

Question. Which versions of the endpoint and sharp-remainder arguments persist near nontrivial roots of unity, where several Borel rays and residue classes interact? The real positive kernel is lost in general. The comparison with the q -Pochhammer literature [4, 5] suggests natural complex variants but not a proof of the positivity-based minimum. A first tractable task is a contour-specific remainder bound away from coalescing rays, before attempting a uniform transition across them.

10.8 Machine-checked inequalities and interval implementation

Question. Can the exact kernel identity, finite geometric remainder, log-convexity, and residual certificate be formalized as a small reusable Lean module, independently of a full transseries datatype? The finite geometric identity and positive-sum estimates are promising first targets. Tonelli interchange, the endpoint Riemann estimate, and the uniform gamma-saddle constants should then be separate theorems. In parallel, an outward-rounded implementation of (74) would turn the present executable checks into numerical certificates. Neither deliverable is included as already completed here.

10.9 Recovering a modular sector from additional functional data

Question. Which minimal functional relations, normalization, and growth conditions determine the flat modular correction uniquely from the Borel-summed carrier? The exact eta identity supplies such data for this example. Abstractly, the same asymptotic power series admits additions of flat functions. A useful uniqueness theorem should state the extra hypotheses explicitly; formal coefficients alone cannot provide them.

11 Conclusion

The main mechanism is elementary but stronger than a fixed-order asymptotic expansion. A positive partial-fraction remainder remains exact when the truncation order grows with the large parameter. That permits a uniform endpoint theorem, a global optimal-truncation argument through moment log-convexity, and a sharp comparison with an independently normalized

modular sector. The resulting transition occurs at $hx = 2\pi$ with a logarithmic displacement of order $(\log x)/x$, and is preserved by exact local inverse transport.

The Borel calculation explains the scale through a stable nearest pole pair, while the remainder calculation fixes the prefactor and the true minimum. The inverse construction then separates a divergent but exactly summable carrier from a genuinely convergent modular correction series. These distinctions are precisely what is needed to extend the inspected repository chapter without conflating a formal all-order formula, an asymptotic error estimate, a Borel sum, and an exact normalized function.

A Source and claim audit

Item	Status in this manuscript
Exact interpolation and phase	Taken from the inspected ProveIt companion; normalization and derivative rederived in section 2.
Eta modular factor	Classical identity, already used in the companion; not a novelty claim.
Fixed-order Bernoulli coefficients	Already in the companion; regularized at the endpoint here.
Uniform $\tau = 0$ matching	Proved in theorem 4.1, addressing the explicit boundary of the companion's fixed- τ_0 result.
General q -product resurgence	Established antecedents [3, 4, 5]; not claimed new.
Central Borel pole cancellations	Explicitly derived here, with signs, residues, and scaled endpoint fixed. No literature-wide priority claim.
Sharp minimum and visibility layer	Full proofs in theorems 6.2 and 7.2; the central quantitative extension advanced for independent checking.
Modular inverse sectors	Proved convergent about the exact carrier, with explicit radii and tails. Does not prove general inverse resurgence.
Numerical verification	Actual run of supplied code; exact symbolic subchecks and 110-digit numerical checks. Not interval-certified.
Lean status	No new Lean file supplied, and no claim of machine-checking these analytic theorems.
Repository coverage	Relevant companion ranges and READMEs inspected; the full oversized canonical TeX was not read.

B Reproduction and file inventory

The package contains `article.tex`, the compiled `article.pdf`, `verify.py`, the actual `verification_results.json`, a short `README.md`, and pinned numerical dependencies in `requirements.txt`. The article is self-contained and does not require the repository's notation file. It can be compiled using

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The PDF was rendered and visually inspected during preparation. The LaTeX source is the editable mathematical record; the numerical script is supplementary evidence and an implementation of the displayed positive sum, not a replacement for the proofs.

References

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